The Teller–Wheeler view prevailed. On 10 March 1950, President Truman ordered a crash program to develop the super.

The Americans' 1949 design for the super appears in retrospect to have been a prescription for failure, just as Oppenheimer's committee had suspected. However, since it was not certain to fail, and since nothing better was known, it was pursued intensely until March 1951, when Teller and Stanislaw Ulam invented a radically new design, one that showed bright promise.

The Teller–Ulam invention at first was just an idea for a design. As Hans Bethe has said, "Nine out of ten of Teller's ideas are useless. He needs men with more judgement, even if they be less gifted, to select the tenth idea, which often is a stroke of genius." To test whether this idea was a stroke of genius or a deceptive dud required turning it into a concrete and detailed bomb design, then carrying out extensive computations on the biggest available computers to see whether the design might work, and then, if the calculations predicted success, constructing and testing an actual bomb.

Two teams were set up to carry out the calculations: One at Los Alamos, the other at Princeton University. John Wheeler led the Princeton team. Wheeler's team worked night and day for several months to develop a full bomb design based on the Teller–Ulam idea, and to test by computer calculations whether it would work. As Wheeler recalls, "We did an immense amount of calculation. We were using the computer facilities of New York, Philadelphia, and Washington—in fact, a very large fraction of the computer capacity of the United States. Larry Wilets, John Toll, Ken Ford, Louis Henyey, Carl Hausman, Dick l'Olivier, and others worked three six-hour stretches each day to get things out."

When the calculations made it clear that the Teller–Ulam idea probably would work, a meeting was called, at the Institute for Advanced Study in Princeton (where Oppenheimer was the director), to present the idea to Oppenheimer's General Advisory Committee and its parent U.S. Atomic Energy Commission. Teller described the idea, and then Wheeler described his team's specific design and its predicted explosion. Wheeler recalls, "While I was starting to give my talk, Ken Ford rushed up to the window from outside, lifted it up, and passed in this big chart. I unrolled it and put it on the wall; it showed the progress of the thermonuclear combustion [as we